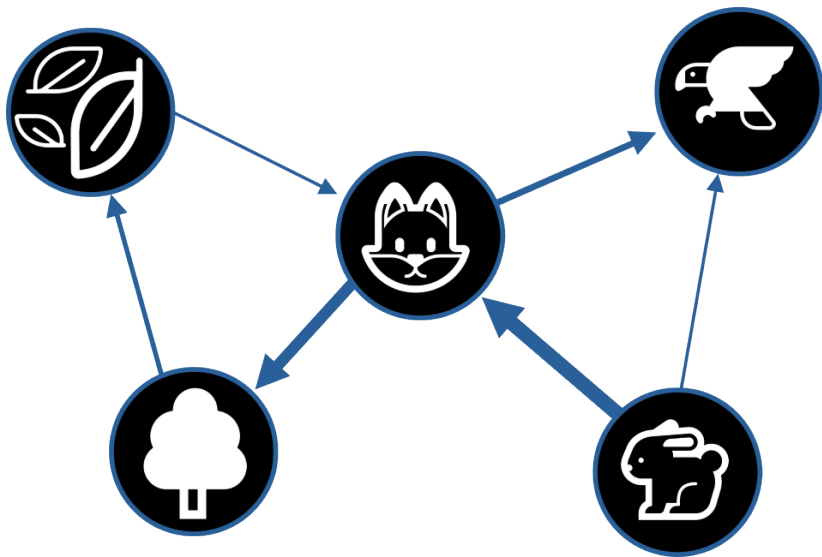


A general tool to create games with valid ecological models for research and stakeholder engagement

Brad Duthie, Lucinda Kirkpatrick, Jianyu Chen, Theodor Cimpeanu, Chloe Corcoran, Conor McKeown, Yuan Pan, Mindi Premarathne, Marly Samuel, Diana Valero, and Nils Bunnefeld

15 June 2026

Modelling to understand biodiversity networks



Human decision-making and Sustainable Development Goals

Ecosystems and how people interact with them (SDGs 3, 7, 13, 15, & 16)

- ▶ Biodiversity loss¹
- ▶ Ecosystem management²
- ▶ Food security³
- ▶ Energy management⁴



¹Mason, T. H. et al. 2018. *Conserv. Lett.* 11:1-9.

²Defries, R & H Nagendra. 2017. *Science* 356:265-270.

³Gould, F, et al. 2018. *Science* 360:728-732.

⁴Thollander, P, et al. 2019. *Sustainability* 11:1-11.

Social simulation games are very popular

Decision-making in complex social-ecological systems is the focus of some of the most popular games of all time¹.

- ▶ **Farmville**

- ▶ 80 million players
- ▶ Diverse player base^{2,3}

- ▶ **SimCity**

- ▶ 1989 to present
- ▶ Millions of copies sold

People invest a lot in these games and take their decisions seriously^{1,4}.



¹Duthie et al. 2021. *Conserv. Biol.* 35:1051-1053.

²ESA. Essential Facts 2019. *Entertainment Software Association*.

³Berry, N. Facebook Casual Game Demographics.

(<http://www.datagenetics.com/blog/december12010/>), accessed 2020-11-17.

⁴Yee, N. 2006. Avatars at work and play. Pp 187-207. Springer.

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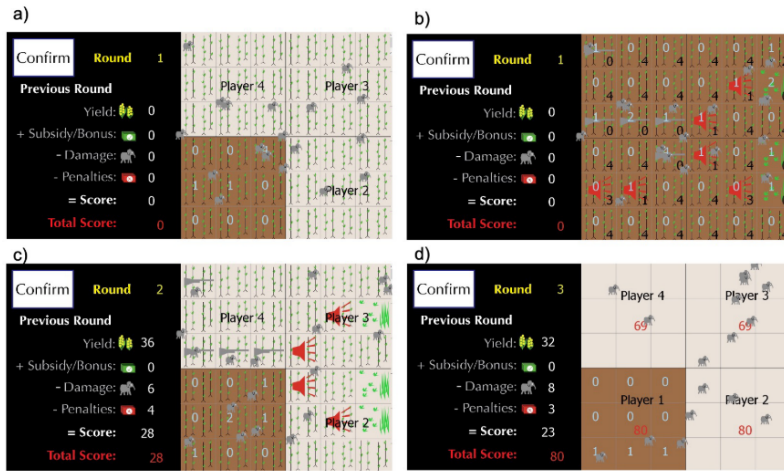
²ESA. Essential Facts 2019. *Entertainment Software Association*.

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⁴Yee, N. 2006. Avatars at work and play. Pp 187-207. Springer.

ConFooBio games (tablet-based and face-to-face)



¹Rakotonarivo, OS, et al. 2021. *Ecol. Soc.* 26:8.

²Rakotonarivo, OS, et al. 2021. *People & Nature* 3:162-175.

³Rakotonarivo, OS, et al. 2021. *Front. Environ. Sci.* 2:661987.

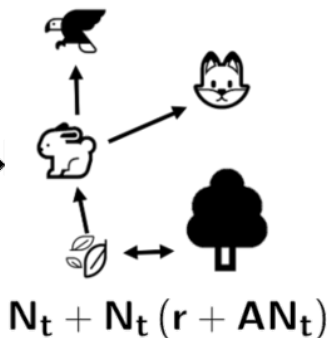
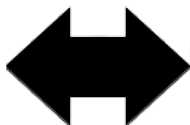
How to interpret stakeholder comments on games?



¹Image: Public Domain

Equivalence between games and models

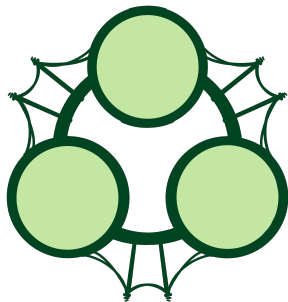
Both games and models must represent how a social-ecological system works.



Conversations about games are about modelling.¹

¹Minderman, J, et al. 2021. *bioRxiv*.

Ecological Knowledge Games Project



A tool to build games with scientifically valid models

- ▶ Highly flexible modelling
- ▶ Custom landscapes & art
- ▶ Engaging game play
- ▶ Big data collection
- ▶ Free and open source

Scientists, artists, & game designers^{1,2}

¹Pan, Y, et al. 2026. Insights for conservation from the Ecological Knowledge Games project. *Conserv. Biol.* e70308

²<https://ecoknowgames.github.io/>

EcoKnowGames Creator tool: Models, art, games



Customise entity icons and colours

¹<https://ecoknowgames.github.io/>

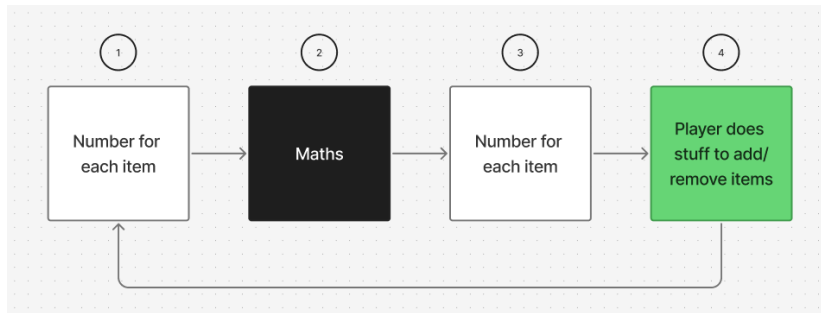
EcoKnowGames Creator tool: Landscape and zones



Customise the landscape and distributions

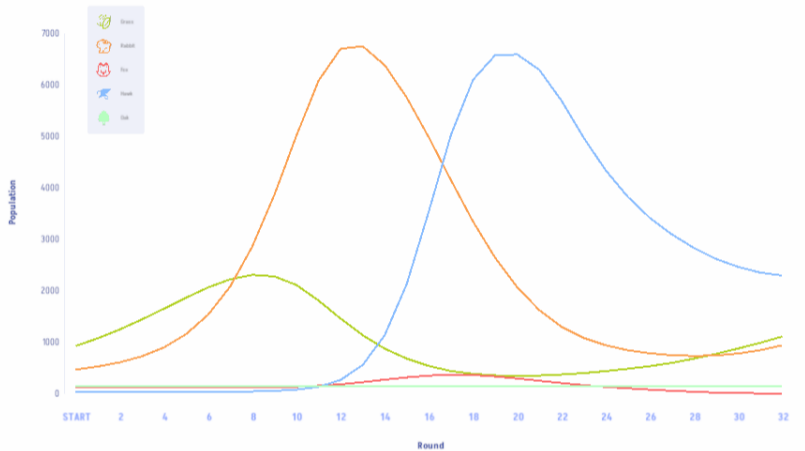
The general EcoKnowGames framework

- ▶ Players make decisions affecting entities
- ▶ Ecological model updates between rounds
- ▶ Players collect items and use currency

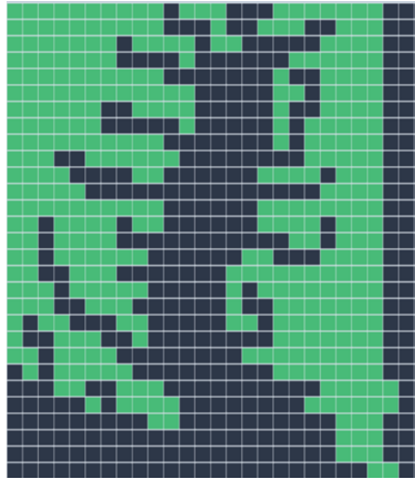
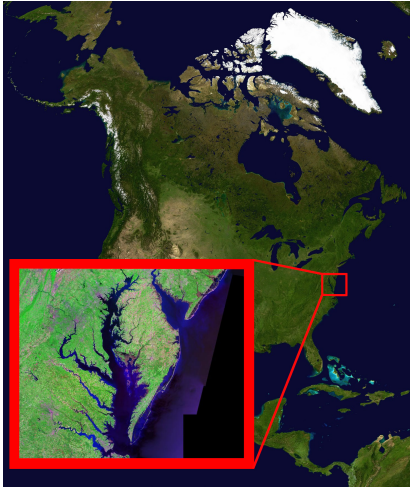


In-game model and player decision data are stored as JSON file.

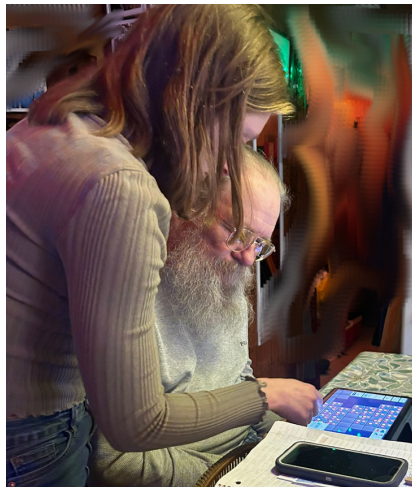
No player intervention is equivalent to a model simulation



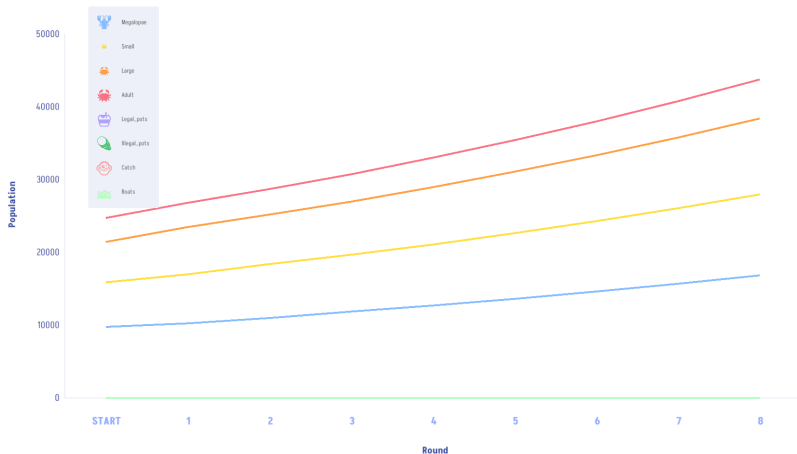
Chesapeake Bay and blue crab fishing



Chesapeake Bay and blue crab fishing



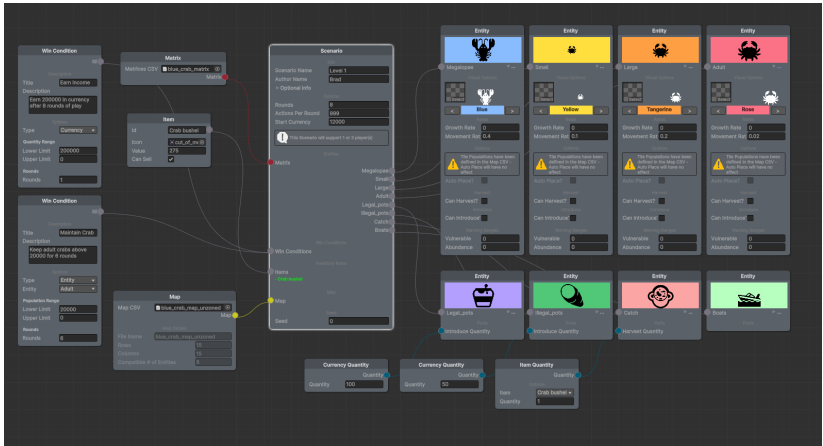
Chesapeake Bay and blue crab fishing



¹Miller, T.J. 2001. *Estuaries*: 535–44.

¹Miller, T.J. 2003. *Bull. Mar. Sci.* 72: 567–88.

EcoKnowGames model-game framework



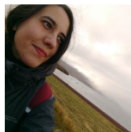
EcoKnowGames Creator: point & click

EcoKnowGames model-game framework



- ▶ Development on GitHub
- ▶ Collaboration welcome
- ▶ Models, art, games, new case studies

Acknowledgements



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UK Research
and Innovation

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